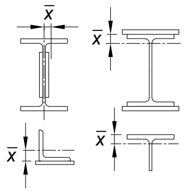
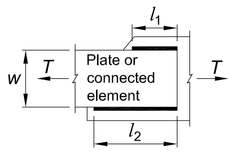
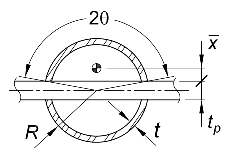
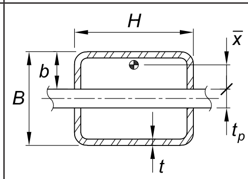
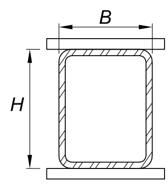
# TABLE D3.1
# Shear Lag Factors for Connections to Tension Members

| Case | Description of Element | Shear Lag Factor, $U$ | Examples |
|---|---|---|---|
| 1 | All tension members where the tension load is transmitted directly to each of the cross-sectional elements by fasteners or welds (except as in Cases 4, 5, and 6). | $U = 1.0$ | – |
| 2 | All tension members, except HSS, where the tension load is transmitted to some but not all of the cross-sectional elements by fasteners or by longitudinal welds in combination with transverse welds. Alternatively, Case 7 is permitted for W, M, S, and HP shapes and Case 8 is permitted for angles. | $U = 1 - \dfrac{\bar{x}}{l}$ |   |
| 3 | All tension members where the tension load is transmitted only by transverse welds to some but not all of the cross-sectional elements. | $U = 1.0$ and $A_n$ = area of the directly connected elements | – |
| 4[a] | Plates, angles, channels with welds at heels, tees, and W-shapes with connected elements, where the tension load is transmitted by longitudinal welds only. See Case 2 for definition of $\bar{x}$. | $U = \dfrac{3l^2}{3l^2 + w^2}\left(1 - \dfrac{\bar{x}}{l}\right)$ |   |
| 5 | Round and rectangular HSS with single concentric gusset through slots in the HSS. | $\bar{x} = \dfrac{R\sin\theta}{\theta} - \dfrac{1}{2}t_p$<br>$\theta$ in rad<br>$U = \left[1 + \left(\dfrac{\bar{x}}{l}\right)^{3.2}\right]^{-10}$ |   |
| | | $\bar{x} = b - \dfrac{2b^2 + tH - 2t^2}{2H + 4b - 4t}$<br>$U = 1 - \dfrac{\bar{x}}{l}$ |   |
| 6 | Rectangular HSS with two side gusset plates. | $U = \dfrac{BU_B + HU_H}{H + B}$<br>$U_B = \dfrac{3l^2}{3l^2 + B^2}$<br>$U_H = \dfrac{3l^2}{3l^2 + H^2}$ |   |

$B$ = overall width of rectangular HSS member, measured 90° to the plane of the connection, in. (mm); $D$ = outside diameter of round HSS, in. (mm); $H$ = overall height of rectangular HSS member, measured in the plane of the connection, in. (mm); $d$ = depth of section, in. (mm); for tees, $d$ = depth of the section from which the tee was cut, in. (mm); $l$ = length of connection, in. (mm); $w$ = width of plate, in. (mm); $\bar{x}$ = eccentricity of connection, in. (mm).

[a] $l = \dfrac{l_1 + l_2}{2}$, where $l_1$ and $l_2$ shall not be less than 4 times the weld size.

**TABLE D3.1 (continued)**
**Shear Lag Factors for Connections to Tension Members**

| Case | Description of Element | | Shear Lag Factor, *U* | Examples |
|---|---|---|---|---|
| 7 | W-, M-, S-, or HP-shapes, or tees cut from these shapes. (If *U* is calculated per Case 2, the larger value is permitted to be used.) | with flange connected with three or more fasteners per line in the direction of loading | $b_f \geq \frac{2}{3}d$, $U = 0.90$<br>$b_f < \frac{2}{3}d$, $U = 0.85$ | – |
| | | with web connected with four or more fasteners per line in the direction of loading | $U = 0.70$ | – |
| 8 | Single and double angles. (If *U* is calculated per Case 2, the larger value is permitted to be used.) | with four or more fasteners per line in the direction of loading | $U = 0.80$ | – |
| | | with three fasteners per line in the direction of loading (with fewer than three fasteners per line in the direction of loading, use Case 2) | $U = 0.60$ | – |

$B$ = overall width of rectangular HSS member, measured 90° to the plane of the connection, in. (mm); $D$ = outside diameter of round HSS, in. (mm); $H$ = overall height of rectangular HSS member, measured in the plane of the connection, in. (mm); $d$ = depth of section, in. (mm); for tees, $d$ = depth of the section from which the tee was cut, in. (mm); $l$ = length of connection, in. (mm); $w$ = width of plate, in. (mm); $\bar{x}$ = eccentricity of connection, in. (mm).

[a] $l = \dfrac{l_1 + l_2}{2}$, where $l_1$ and $l_2$ shall not be less than 4 times the weld size.

## D5. PIN-CONNECTED MEMBERS

### 1. Tensile Strength

The design tensile strength, $\phi_t P_n$, and the allowable tensile strength, $P_n/\Omega_t$, of pin-connected members, shall be the lower value determined according to the limit states of tensile rupture, shear rupture, bearing, and yielding.

(a) For tensile rupture

$$P_n = F_u(2tb_e) \tag{D5-1}$$

$$\phi_t = 0.75 \text{ (LRFD)} \qquad \Omega_t = 2.00 \text{ (ASD)}$$

(b) For shear rupture

$$P_n = 0.6C_r F_u A_{sf} \tag{D5-2}$$

$$\phi_{sf} = 0.75 \text{ (LRFD)} \qquad \Omega_{sf} = 2.00 \text{ (ASD)}$$

where

$A_{sf} = 2t(a + d/2)$
= area on the shear failure path, in.$^2$ (mm$^2$)